

1

Dataset

BanglishRev

Daraz e-commerce reviews
119,859 reviews
Bangla, English, Romanized,
and code-mixed text

– 11 corrupted rows (#NAME?)
→ 119,848 reviews

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Labels & groups

Star rating → label

1-3★ → negative (17.7%)
4-5★ → positive (82.3%)

Script-based groups:

- Bangla script 36.1%
- English 37.9%
- Romanized Bangla 5.7%
- Romanized + Eng. 15.5%
- Mixed script 4.7%

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Preprocessing

A. Minimal cleaning

NFC, strip HTML
URL → <URL>, @user → <USER>
keep hashtags, emojis,
negation, English words

B. Normalized Banglish

A + lowercase, Bangla digits,
elongation (sooo → soo),
138 spelling variants → 67 forms
(vlo, bhalo → valo)

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Data split

Stratified 80 / 20

stratified by label × group
seed 42, same for all models

Train 86,290 (72%)
Validation 9,588 (8%)
Test 23,970 (20%)

validation: tuning &
early stopping only

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Classifiers

TF-IDF baselines

word 1-2 or char 2-5-grams
× LR or linear SVM
class_weight = balanced
C tuned on validation

Fine-tuned transformers

XLNet-base (3 seeds)
BanglaBERT (1 seed)
own tokenizer, ≤ 128 tokens
weighted CE, best epoch on val.

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Evaluation

20% test set

Macro-F1 (primary)
per-class P / R / F1
confusion matrix
accuracy (supplementary)

per language group
mean ± std over seeds
paired bootstrap
(1,000 resamples)

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Analysis

What we examine

Model comparison
per language group

Normalization effect
(B – A) per group

Tokenizer fertility

Error analysis &
rating-label noise

every model is trained and tested on both A and B

same split, same metrics for all runs